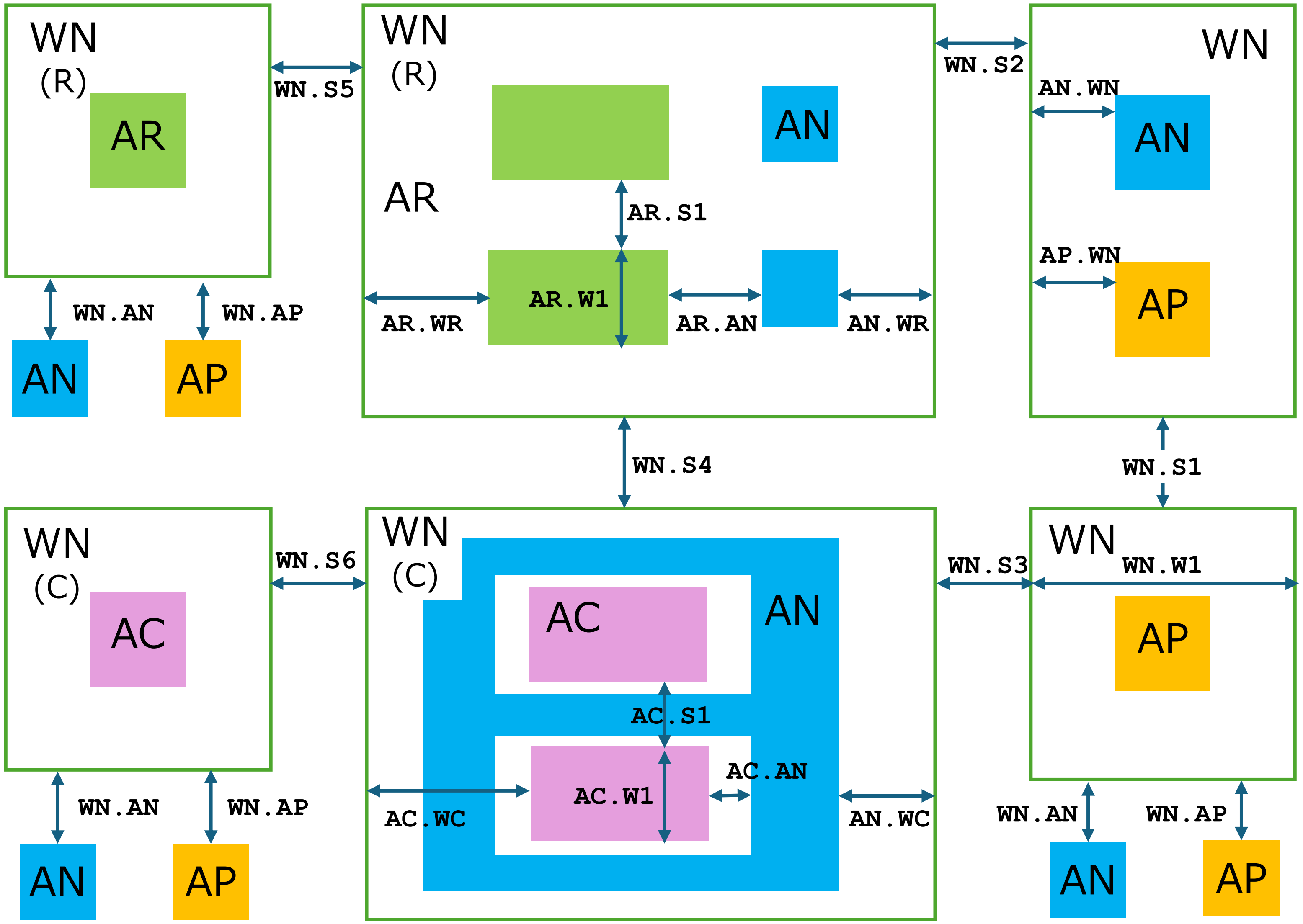
Design Rules

for Drawing Layers

Drawing Layer Table

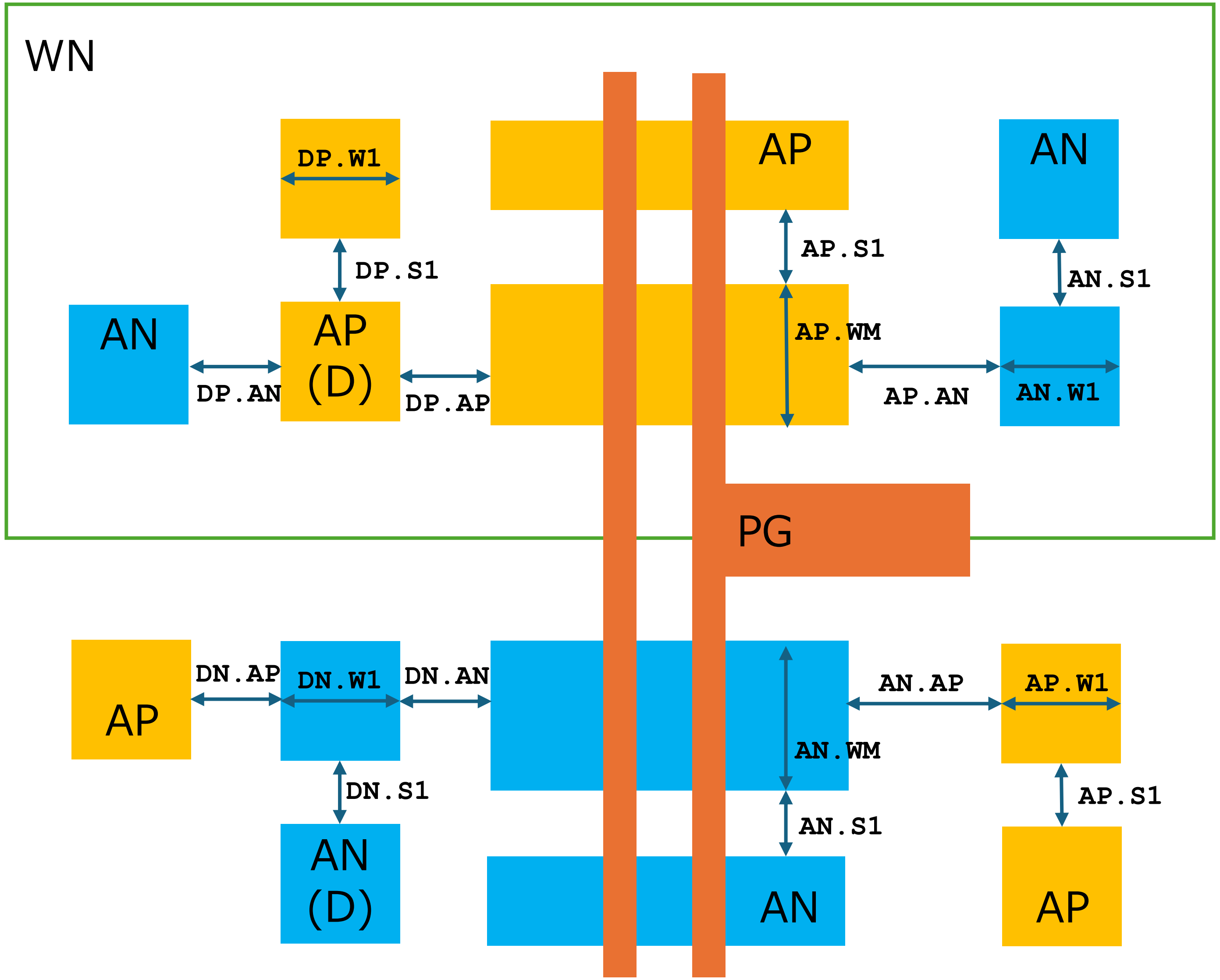
	Original MASK layers			Proposed Drawing layers			
	Name	NUM	TYPE	Name	NUM	TYPE	
1	PSUB	140	0	WN	140	0	
2	NW	36	0				MDP generatioed
3	HVNW	141	0				MDP generatioed
4	L	3	0				MDP generatioed
				AP	3	1	Active for P+
				AN	3	2	Active for N+
				AR	3	3	Active for resistor
				AC	3	4	Active for capacitance
5	NF	25	0				MDP generatioed
6	PF	26	0				MDP generatioed
7	CL	143	0				MDP generatioed
8	HPBE	144	0				No USE
9	HNBE	145	0				No USE
10	PBE	146	0				MDP generatioed
11	NBE	147	0				MDP generatioed
12	HPM	33	0				MDP generatioed
13	RHP	148	0				No USE
14	SG	8	0				
				PO	8	1	POLY transistor
				PR	8	2	POLY resistor
15	PM	35	0				MDP generatioed
16	NM	7	0				MDP generatioed
17	R	12	0				MDP generatioed
18	PSD	9	0				MDP generatioed
19	NSD	28	0				MDP generatioed
20	CONT	11	0	CO	11	0	
21	M1	13	0	M1	13	0	
22	TC	19	0	V1	19	0	
23	M2	20	0	M2	20	0	
24	PRO	14	0	PE	14	0	
	TXM1	48	0	TXM1	48	0	M1 Label
	PIN	48	1	PIN	48	1	M1 Pin
	TXM2	49	0	TXM2	49	0	M2 Label
	MASK	63	0	MASK	63	0	Waiver region
	MEAS	63	1	MEAS	63	1	Measurement region
	SCRIBE	80	0	SCRIBE	80	0	Waiver region (EDGE SEAL)
	prBoundary	235	0	prBoundary	235	0	Cell boundary

WN to AP/AN/AR/AC related



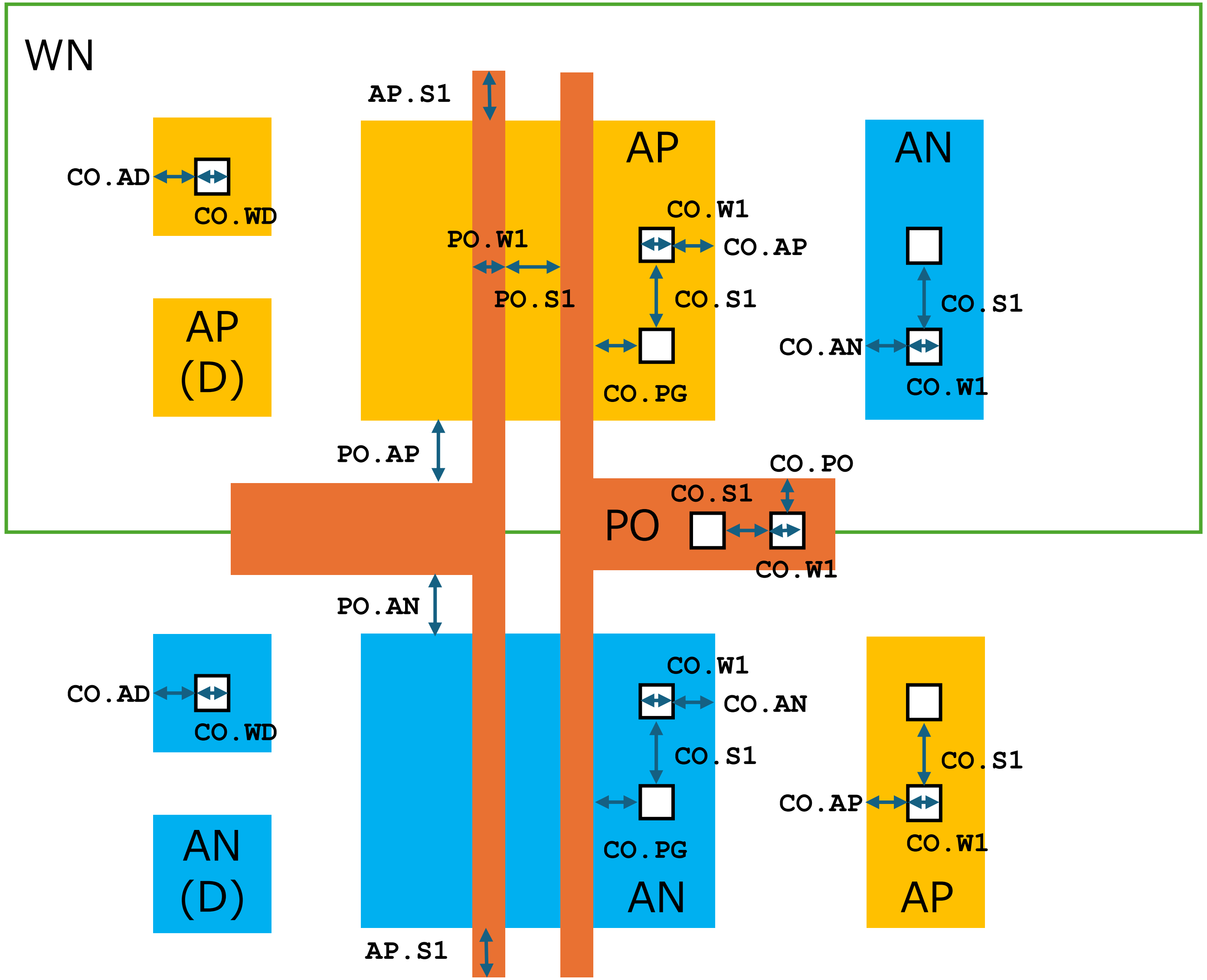
Name	Description	Rule
WN.W1	WN Wmin	8.0
WN.S1	WN-WN Smin	12.0
WN.S2	WN-WN(R) Smin	9.5
WN.S3	WN-WN(C) Smin	12.0
WN.S4	WN(R)-WN(C) Smin	9.5
WN.S5	WN(R)-WN(R) Smin	8.0
WN.S6	WN(C)-WN(C) Smin	12.0
WN.AP	WN-AP Smin	10.0
WN.AN	WN-AN Smin	5.0
AR.W1	AR Wmin	2.8
AR.W2	AR Wmax	20.0
AR.L1	AR Lmin	13.0
AR.L2	AR Lmax	100.0
AR.S1	AR Smin	4.0
AR.WR	AR-WN(R) Emin	10.0
AN.WR	AN-WN(R) Emin	5.0
AR.AN	AR-AN Smin	4.0
AC.W1	AC Wmin	28.5
AC.W1	AC Wmax	120.0
AC.S1	AC Smin	???
AC.WC	AC-WN(C) Emin	???
AN.WC	AN-WN(C) Emin	3.0
AC.AN	AC-AN Smin	2.8
AC.R1	AN must surround AC	

MOS transistor related



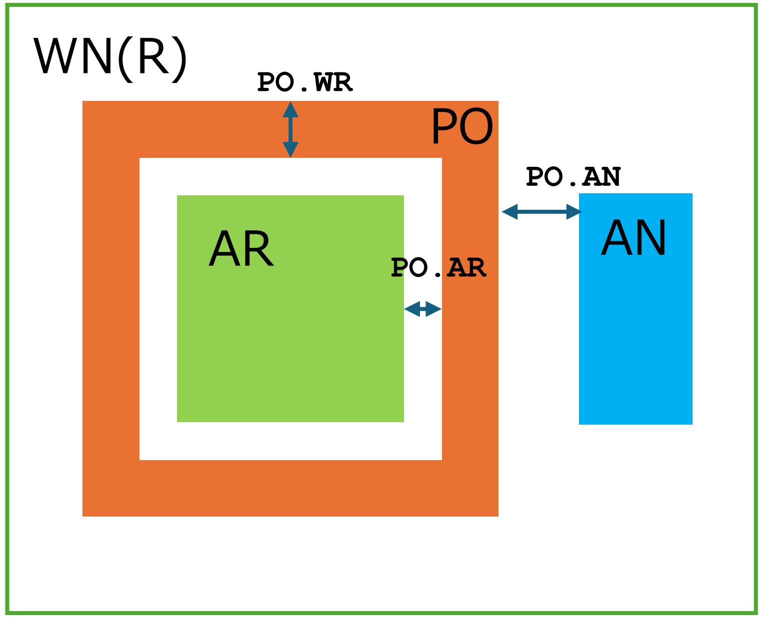
Name	Description	Rule
AP.W1	AP Wmin	1.4
AP.S1	AP Smin	1.4
AP.AN	AP-AN Smin	2.8
AP.WM	PMOS Wmin	3.4
AP.WM	PMOS Wmax	60.0
AP.WN	AP-WN Emin	7.0
DP.W1	AP Wmin	1.4
DP.S1	DP Smin	???
DP.AP	DP-AP Smin	2.8
DP.AN	DP-AN Smin	2.8
AN.W1	AN Wmin	1.4
AN.S1	AN Smin	1.4
AN.AP	AN-AP Smin	2.8
AN.WM	NMOS Wmin	3.4
AN.WM	NMOS Wmax	60.0
AN.WN	AN-WN Emin	5.0
DN.S1	DN Smin	-
DN.AP	DP-AP Smin	2.8
DN.AN	DP-AN Smin	2.8

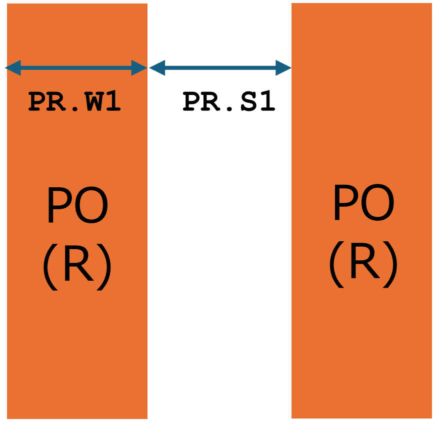
Contact and Poly related

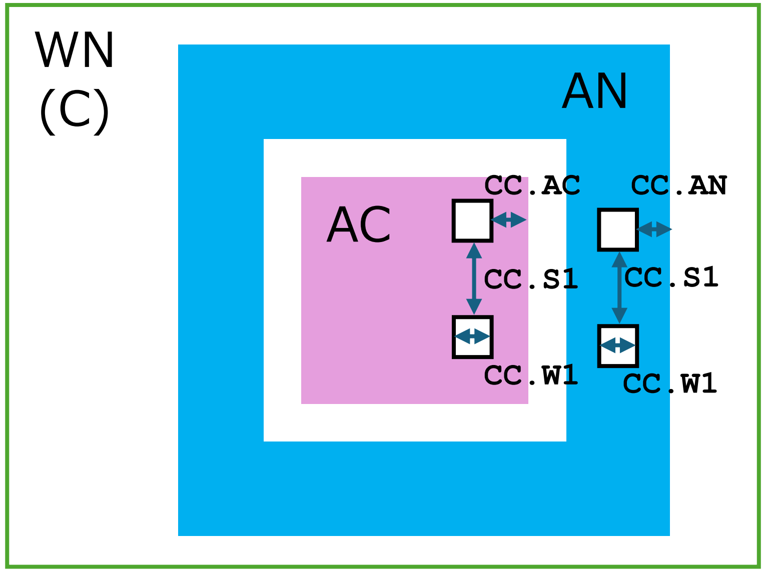


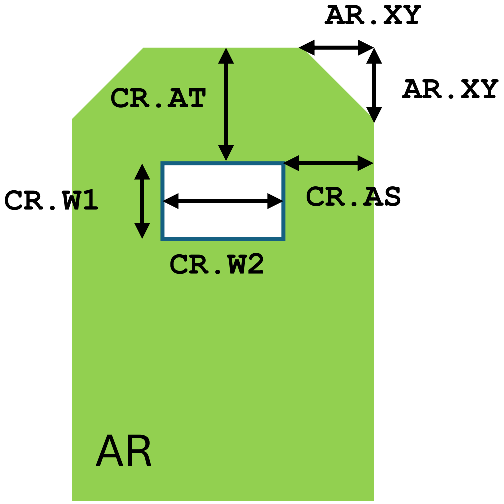
Name	Description	Rule
PO.W1	PO Wmin	1.0
PO.W1	PO Wmax	30.0
PO.S1	PO Smin	1.2
PO.AP	PO-AP Smin	0.4
PO.AN	PO-AN Smin	0.4
PO.EM	MOS Endcap min	1.2
CO.W1	CO Wmin	1.0
CO.S1	CO Smin	1.0
CO.WD	CO(D) Width	1.2
CO.AP	CO-AP Emin	0.8
CO.AN	CO-AN Emin	0.8
CO.PG	CO-PO(G) Smin	1.0
CO.PO	CO-PO Emin	0.8
CO.AD	CO-AD Emin	1.2

Resistor(AR/AS) and Capacitor(CSIO) related



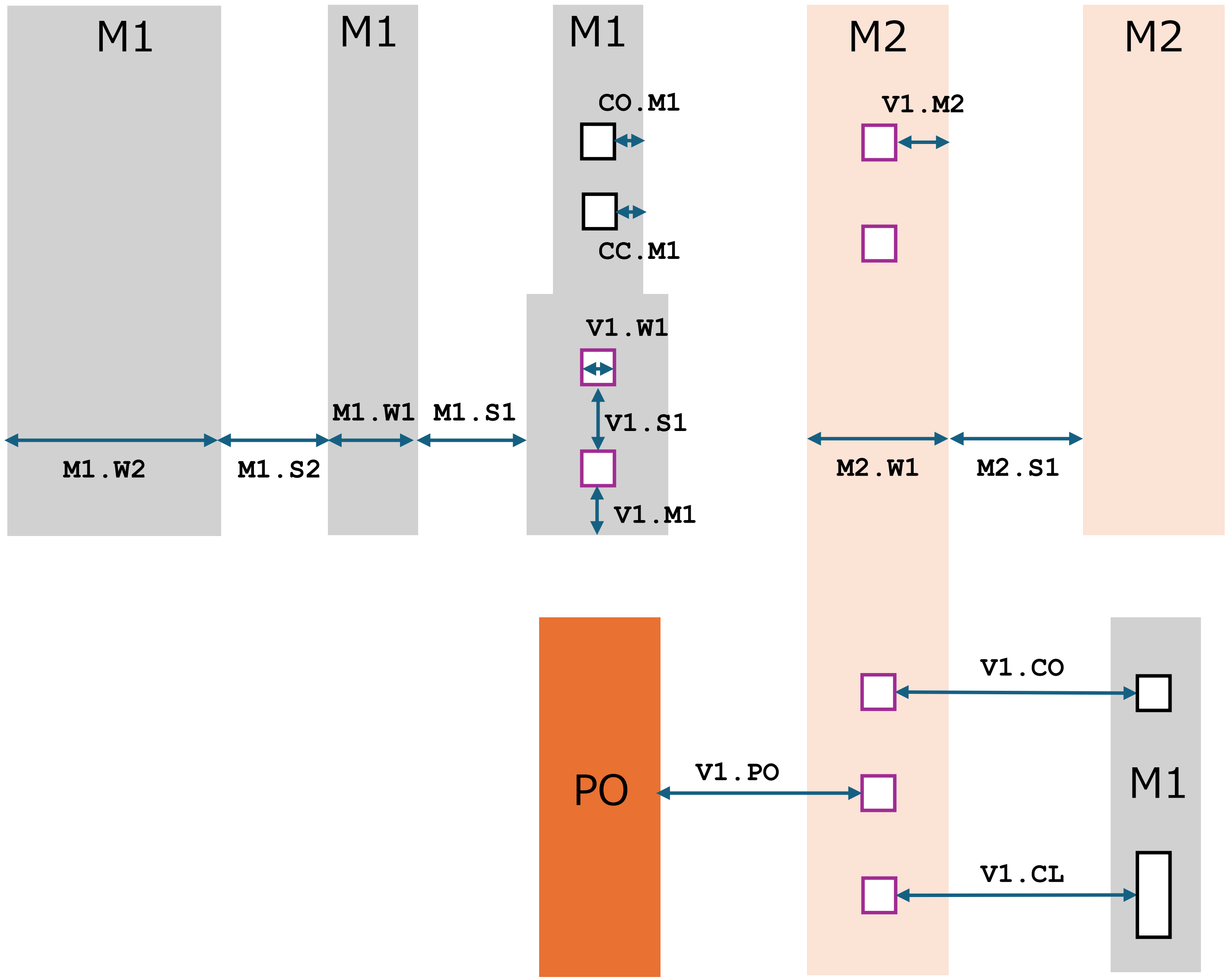






Name	Description	Rule
PO.AR	PO-AR Smin	1.0
PO.WR	PO(AR) Wmin	2.0
PO.AN	PO(AR)-AN Smin	???
PO.R1	PO must surround AR	
PO.R2	PO must tie-down	
PR.W1	PO(R) Wmin	2.8
PR.W1	PO(R) Wmin	4.0
PR.W1	PO(R) Wmax	20.0
PR.L1	PO(R) Lmin	20.0
PR.L1	PO(R) Lmax	100.0
PR.S1	PO(R) Smin	2.0
AR.XT	AR corner cut	1.0
CR.W1	CO(RR) Width	1.0
CR.W2	CO(RR) Wmin	1.0
CR.AT	CO(RR)-AR(Top) Emin	1.5
CR.AS	CO(RR)-AR(Side) Emin	1.0
CC.W1	CO(C) Wmin	1.2
CC.S1	CO(C) Smin	1.2
CC.AC	CO(C)-AC Emin	2.9
CC.AN	CO(C)-AN Emim	1.2
CO.M1	CO-M1 Emin	1.0
CC.M1	CO(C)-M1 Emin	1.2

M1/V1/M2 related



Name	Description	Rule
M1.W1	M1 Wmin	1.8
M1.S1	M1 Smin	1.4
M1.W1	M1(W) Wmin	10.0
M1.S2	M1(W) Smin	2.0
V1.W1	V1 Windth	1.4
V1.S1	V1 Smin	1.5
V1.M1	V1-M1 Emin	1.0
V1.M2	V1-M2 Emin	1.0
V1.PO	V1-PO Smin	1.2
V1.CO	V1-CO Smin	1.0
V1.CL	V1-CO(L) Smin	1.4
M2.W1	M2 Wmin	3.0
M2.W2	M2 Smin	2.0

質問・確認事項

2025/12/25

DIODE

リファレンス・マニュアルTable I-1-1によればダイオードはP/NどちらもPM+NMインプラと記載

DP：PM+NM
DN：PM+NM

リファレンス・マニュアル図6.1/6.2にも同様の指示あるが、

リファレンス・マニュアル図6.12/6.13に(ESD)では

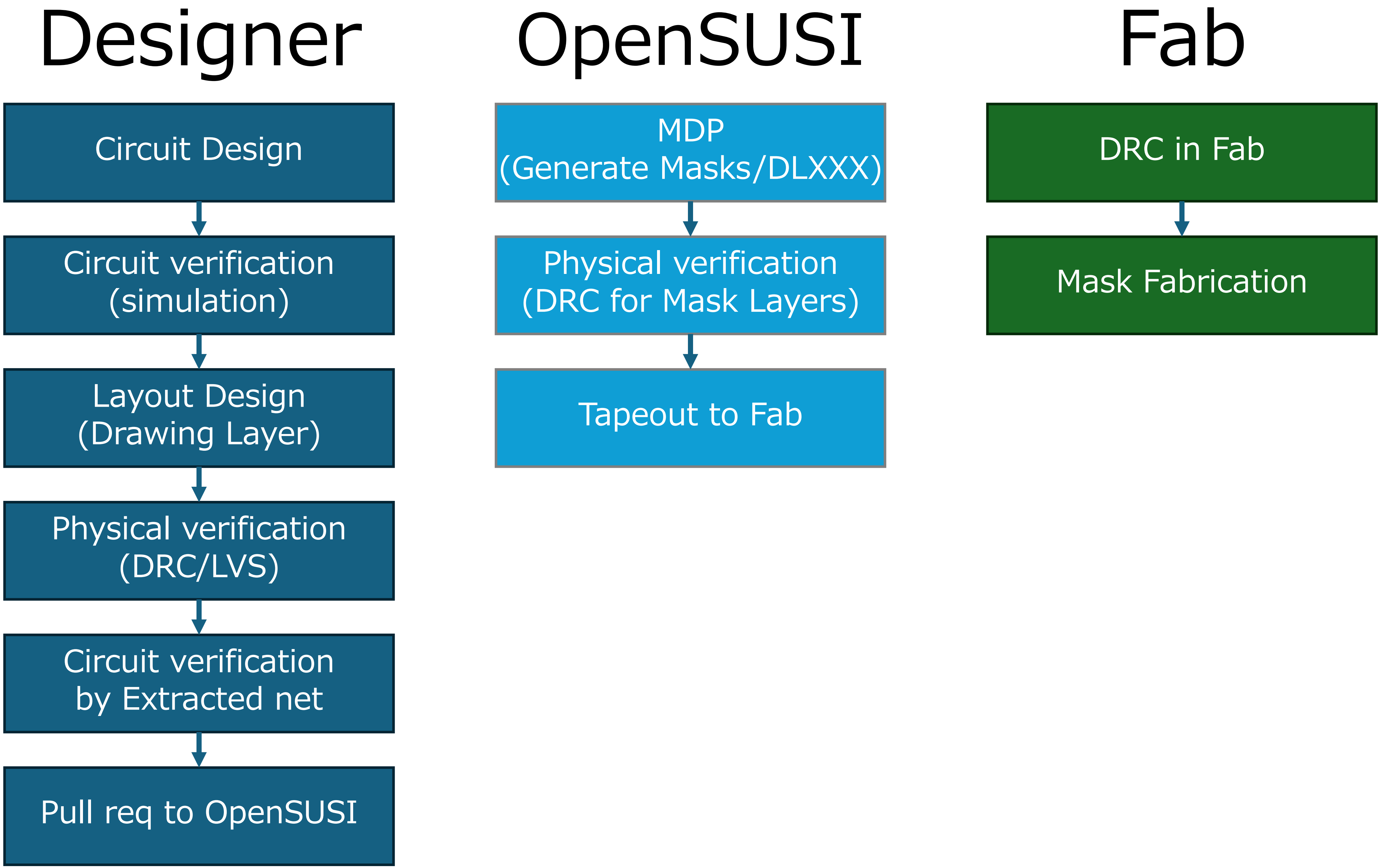
DP：NM
DN：PM

と指示がある。本当はどちらでしょう？違う場合は、Spiceモデルを公開頂けますと幸いです。

また、ESD用NMOSのSPICEモデルも公開頂けますと幸いです。

Design to Tapeout Process

Mask Data Preparation


Designer
Circuit Design
Circuit verification (simulation)
Layout Design (Drawing Layer)
Physical verification (DRC/LVS)
Circuit verification by Extracted net
Pull req to OpenSUSI
OpenSUSI
MDP (Generate Masks/DLXXX)
Physical verification (DRC for Mask Layers)
Tapeout to Fab
Fab
DRC in Fab
Mask Fabrication